



Pre-Registration

Bayesian Reanalysis of Computationally Reproduced Psychology Learning and Teaching Results

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1 Study Information

1. Title Bayesian Reanalysis in Computationally Reproduced Psychology Learning and Teaching Research

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3. Objective For a long period of time, the frequentist null-hypothesis significance testing has been dominant in Psychology. Therefore we propose to use Bayesian analysis to re-examine the existing studies in Psychology Learning and Teaching.

In detail, the project will conduct a Bayesian reanalysis of statistical results associated with the main article claims, previously identified and computationally reproduced in the project *The Computational Reproducibility of Psychology Learning and Teaching Research* by Almuhanha and Allen (2026).

The first objective is to assess how often the inferential interpretation of each frequentist result, based on the decision rule used in the original study, agrees with the conclusion drawn from the Bayesian evidence, and how often they diverge. The second objective is to investigate which statistical and design features are associated with these patterns. The final objective is to evaluate when Bayesian analysis adds useful information in this subfield, especially by distinguishing evidence favouring a null model from evidence that remains inconclusive.

4. Research Questions

1. How often do reproduced frequentist results and Bayesian analyses lead to the same conclusion?
2. Among statistically significant results (favouring the alternative hypothesis), how often does the Bayesian analysis provide clear evidence ($BF_{10} \geq 3$) for the alternative hypothesis, how often does it remain inconclusive ($1/3 < BF_{10} < 3$), and how often does it favour the null hypothesis ($BF_{10} \leq 1/3$)?
3. Among non-significant results (favouring the null hypothesis), how often does the Bayesian analysis provide evidence for the null hypothesis?

esis ($BF_{10} \leq 1/3$), and how often does it remain inconclusive ($1/3 < BF_{10} < 3$)?

4. Within comparable analysis types, to what extent are smaller p-values associated with the larger BF_{10} .
5. How sensitive are BF_{10} values, and the evidence classifications derived from them to reasonable alternative prior specifications?
6. Exploratorily, which statistical and design features lead to the concordance or discordance between the frequentist interpretation and Bayesian evidence classification?
7. Under what conditions does Bayesian analysis provide useful information beyond the reproduced frequentist result?

2 Design Plan

5. Study type Observational Meta-Research Study

6. Study design

1. The study uses the statistical results identified and successfully reproduced in *The Computational Reproducibility of Psychology Learning and Teaching Research* by Almuhanha and Allen (2026). The present study will not repeat the original claim identification process.
2. Each target result will be linked to the scientific claim it supports, the result reported in the article, the reproduced result, and the data or summary statistics needed for Bayesian analysis.
3. Almuhanha's original scripts used to reproduce the target results in the source study will be reimplemented in R with the assistance of ChatGPT. The R implementations will then be run on the original datasets to obtain the unrounded results, which will be compared to Almuhanha's reproduced results for validation.
4. Each target will be given a clear status. A target may be ready for Bayesian analysis, require more information, remain unresolved, or be deferred because it requires a specialised model.
5. Each eligible target will be analysed using a Bayesian model that reflects the original study design and the quantity that the original analysis aimed to test.
6. The core project will be completed in two waves:
 - (a) Wave 1 will include simple analyses that can be represented by one main effect or one correlation. These are one-sample t-tests, paired-samples t-tests with exactly two conditions or time points, equal-variance independent-groups t-tests with two groups, and Pearson correlations. Two-group ANOVAs will also be included in Wave 1 when it is exactly equivalent to one of these t-tests, with the same sample, contrast, error term, and variance assumptions.
 - (b) Wave 2 will include more complex ANOVA and regression models. This includes one-way ANOVAs with more than two groups, factorial ANOVAs, interactions, models allowing unequal group

variances, repeated-measures ANOVAs with more than two conditions or time points, within-subject factorial designs, mixed designs, and multilevel models.

7. Wave 3 will contain specialised multivariate, structural-equation, rank-based, and ordinal analyses. These analyses are planned extensions. They will only be added when the data, model, priors, and comparison can be specified clearly and the model passes the required checks.
8. For each model family, the null model, alternative model, main prior, and sensitivity priors will be specified before the final Bayesian result is produced.
9. The Wave 1 model specifications will include the default JZS Bayes factor, BF_{10} , under the primary prior: a Cauchy prior on the standardized effect with scale $r = 0.707$ for t -tests and $r = 1.0$ for Pearson correlations (Rouder et al., 2009; Ly et al., 2016). BF_{10} will also be calculated under two further scales for the sensitivity analysis: $r \in \{0.5, 0.707, 1.0\}$ for t -tests and $r \in \{0.5, 1.0, 1.5\}$ for correlations. The prior-sensitivity span will be calculated as $S_i = \max_k \log_{10}(BF_{10,ik}) - \min_k \log_{10}(BF_{10,ik})$, together with indicators of whether the evidence category and the favoured hypothesis change across those priors. The evidence category (Table 1) and the six-cell concordance classification will also be reported, together with $-\log_{10}(p)$ and $\log_{10}(BF_{10})$.
10. Frequentist–Bayesian concordance will be examined using both categories and continuous values.
11. R will be the main analysis environment. JASP may be used as an independent check for analyses that can be reproduced in both programs.

3 Sampling Plan

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| 7. Existing data (required) | Registration prior to analysis of the data. |
| 8. Explanation of existing data (optional) | The data used for the study is the frequentist statistical results of existing studies in <i>Psychology Teaching and Learning</i> . By the submission of the Preregistration, the data exists and had been accessed and reviewed by the research team, but no further relevant analysis has been applied. We have not calculated the Bayes factor, compared p-values and Bayes factors, and observed any agreement/disagreement pattern, direction, nor features. |
| 9. Data collection procedures (required) | The data used in the present study are drawn from Ali Almuhanha and Dr Peter Allen’s (2026) study, <i>The Computational Reproducibility of Psychology Learning and Teaching Research</i> . Briefly, the original study randomly sampled 60 of 103 papers identified as having open data and as potentially computationally reproducible. These 103 papers were drawn from 1,039 empirical papers published in Teaching of Psychology (ToP), Psychology Learning and Teaching (PLaT), and Scholarship of Teaching and Learning in Psychology (SoTL-P) between January 1, 2015, and May 29, 2025. For each sampled paper, the source-study researchers identified a central claim and selected its statistical result. When multiple statistical results support a same claim, the result with the largest effect size was selected as the |

primary target result, and an additionally secondary target was randomly selected from the other supporting statistical results.

The main unit of our analysis is a target statistical result linked to a scientific claim. Some papers contribute more than one target result, so results from the same paper will not be treated as fully independent: Descriptive summaries will be reported at the target level, alongside the number of unique papers, and this dependence will be considered when interpreting the results. To construct our dataset, the research team screened papers selected by the source study and identified target results eligible for inclusion in the present Bayesian reanalysis based on the following criteria:

1. The target result was successfully reproduced in the source study
2. A p-value was available for the target result.
3. Sufficient data or summary statistics were available to conduct the prespecified Bayesian analysis.

All targets will remain in the project registry. Targets that cannot be analysed will be labelled as held, deferred, or not assessable rather than silently removed.

10. Sample size (required)

The candidate pool currently contains 47 target statistical results from 29 papers: 27 primary target results and 20 secondary target results.

All 47 targets will remain in the project registry. The core Bayesian sample will include eligible targets from Waves 1 and 2. Targets requiring specialised Wave 3 models may be analysed later as extensions.

The final number of completed Bayesian analyses will therefore depend on the pre-specified eligibility and modelling requirements, not on the observed p -values, Bayes factors, or whether the two approaches agree. The numbers of completed, deferred, held, and not assessable targets will all be reported.

4 Variables

11. Measured variables (required)

The main unit of analysis is a target statistical result linked to a scientific claim.

For each target, if applicable, we will record the reproduced p-value, test statistic, degrees of freedom, analysis sample size, effect estimate, standard error or confidence interval, statistical test type, and whether the original analysis was one-sided or two-sided, and the directionality of the original alternative hypothesis where applicable. Original test sidedness will be coded as one-sided, two-sided, omnibus (for F , χ^2 , and other non-directional omnibus tests), or not applicable. Hypothesis directionality will be coded as directional (stored as positive), non-directional (two-sided), or unclear.

For the Bayesian reanalysis, we will record the model and prior specification, the primary BF_{10} , the posterior effect estimate and credible interval where available, and whether the conclusion changes across the planned sensitivity priors.

The concordance analysis will use the frequentist significance category, the Bayesian evidence category, and the continuous measures

$$-\log_{10}(p) \quad \text{and} \quad \log_{10}(BF_{10}).$$

Possible explanatory variables will include sample size, effect magnitude, uncertainty, analysis type, directionality, prior sensitivity, study design, data-input type, claim role, and study identifier.

12. Indices (optional)

Frequentist results will be classified as significant ($p < .05$) or non-significant ($p \geq .05$), while exact p -values will also be retained. The Bayes factor is

$$BF_{10} = \frac{p(\mathbf{y} | H_1)}{p(\mathbf{y} | H_0)}, \quad BF_{01} = BF_{10}^{-1}.$$

Thus, $BF_{10} > 1$ favours H_1 , $BF_{10} < 1$ favours H_0 , and $BF_{10} = 1$ indicates equal predictive support.

The following table will be used in Bayes factor interpretation

Table 1: Interpretive categories for BF_{10} , adapted from Wetzels et al. (2011), based on Jeffreys (1961).

BF_{10}	Interpretation
≥ 100	Decisive evidence for H_1
30 to < 100	Very strong evidence for H_1
10 to < 30	Strong evidence for H_1
3 to < 10	Substantial evidence for H_1
1 to < 3	Anecdotal evidence for H_1
$= 1$	Equal support for H_0 and H_1
$> 1/3$ to < 1	Anecdotal evidence for H_0
$> 1/10$ to $\leq 1/3$	Substantial evidence for H_0
$> 1/30$ to $\leq 1/10$	Strong evidence for H_0
$> 1/100$ to $\leq 1/30$	Very strong evidence for H_0
$\leq 1/100$	Decisive evidence for H_0

Note. The numerical thresholds are adapted from Wetzels et al. (2011), who based their classification on Jeffreys (1961).

For the main concordance analysis, these categories will be collapsed into evidence favouring H_1 ($BF_{10} \geq 3$), weak or inconclusive evidence ($1/3 < BF_{10} < 3$), and evidence favouring H_0 ($BF_{10} \leq 1/3$).

For RQ2, we will report the proportion of results classified as statistically significant in original studies that fall into each of these three Bayesian evidence categories. For RQ3, we will report the corresponding proportions among results classified as non-significant.

Crossing these with the frequentist classification produces six outcomes: significant with evidence for H_1 , inconclusive evidence, or evidence for H_0 ; and non-significant with evidence for H_0 , inconclusive evidence, or evidence for H_1 .

Continuous comparisons will use

$$X_i = -\log_{10}(p_i), \quad Y_i = \log_{10}(BF_{10,i}).$$

Prior sensitivity will be assessed from the range

$$S_i = \max_k \log_{10}(BF_{10,ik}) - \min_k \log_{10}(BF_{10,ik}),$$

and by whether the favoured hypothesis or evidence category changes across the planned priors. Exact p -values and Bayes factors will remain the primary reported quantities.

5 Analysis Plan

13. Statistical models (required)

Each eligible result will be analysed using a Bayesian model suited to the original study design and research question. Simpler models will be fitted

first, followed by more complex regression, factorial, and repeated-measures analyses. Specialised multivariate, structural-equation, rank-based, and ordinal models will be treated as optional extensions and reported separately. The analyses will be organised into the three waves defined in the Design Plan. Wave 1 will include simple analyses that can be represented by one main effect or one correlation. These are one-sample t-tests, paired-samples t-tests with exactly two conditions or time points, equal-variance independent-groups t-tests with two groups, and Pearson correlations. A two-group ANOVA may also be included in Wave 1 when it is exactly equivalent to one of these t-tests, with the same sample, contrast, error term, and variance assumptions. Wave 2 will include more complex ANOVA and regression models. This includes one-way ANOVAs with more than two groups, factorial ANOVAs, interactions, models allowing unequal group variances, repeated-measures ANOVAs with more than two conditions or time points, within-subject factorial designs, mixed designs, and multilevel models. Wave 3 will contain specialised multivariate, structural-equation, rank-based, ordinal, and other non-standard analyses. These are planned extensions rather than required parts of the core project. They will only be added when the original model can be identified, the necessary data are available, the priors and comparison can be specified clearly, and the implementation passes the required checks.

Frequentist and Bayesian conclusions will be compared using the six concordance categories defined above. We will report counts, proportions, and study-weighted proportions.

The relationship between p -values and Bayes factors will also be examined using graphical summaries and Spearman's rank correlation. Possible explanations for agreement and disagreement will be explored descriptively and, if the final sample is sufficiently large, with a simple regression model.

14. Inference criteria (optional)

Exact p -values, Bayes factors, posterior estimates, and uncertainty intervals will be reported. Thresholds will be used to organise the comparison and will not be treated as fixed boundaries between true and false conclusions. The main frequentist threshold will be

$$p < .05.$$

The main Bayesian classification will use

$$BF_{10} \geq 3$$

for evidence favouring H_1 ,

$$BF_{10} \leq 1/3$$

for evidence favouring H_0 , and values between these limits for weak or inconclusive evidence. Sidedness will follow the original reported analysis. A result the original study reported with as a one-sided test will be evaluated with a one-sided (order-restricted) Bayes factor in the predicted direction and compared to the one-sided p ; all other results will be two-sided.

For models fitted using simulation methods, results will only be treated as complete when the relevant checks are satisfactory. These checks will include convergence, effective sample size, sampling warnings, and whether data generated by the fitted model are reasonably similar to the observed data.

All eligible results will be reported whether they favour H_0 , favour H_1 , or remain inconclusive.

15. Data

Targets will not be excluded because of their p -value, Bayes factor, effect

**exclusion
(optional)**

direction, or whether the two approaches agree.

A target may be assigned a held, deferred, or not-assessable status when:

1. the published result cannot be reproduced or identified reliably;
2. the required raw data or summary information are unavailable;
3. a defensible Bayesian model cannot be specified;
4. it duplicates another target;
5. it is a robustness analysis rather than a separate scientific target.

Held, deferred, and not assessable targets will remain visible in the project registry and coverage tables with a written reason.

**16. Exploratory
analysis
(optional)**

Exploratory analyses may include:

1. detailed examination of targets showing clear frequentist–Bayesian disagreement;
2. comparisons across analysis types and study designs;
3. alternative Bayes-factor thresholds;
4. additional reasonable prior choices;
5. robust or alternative versions of the original models;
6. simulations examining how sample size, true effect size, and prior width affect concordance;
7. any completed Wave 3 analyses.

References

- [1] Almuhan, A., & Allen, P. J. (2026). The computational reproducibility of psychology learning and teaching research. *Teaching of Psychology*. Advance online publication. <https://doi.org/10.1177/00986283251412796>
- [2] Jeffreys, H. (1961). *Theory of Probability* (3rd ed.). Oxford, UK: Oxford University Press.
- [3] Wetzels, R., Matzke, D., Lee, M. D., Rouder, J. N., Iverson, G. J., & Wagenmakers, E.-J. (2011). Statistical evidence in experimental psychology: An empirical comparison using 855 t tests. *Perspectives on Psychological Science*, 6(3), 291–298. <https://doi.org/10.1177/1745691611406923>